Ans. (b) : Required numbers

3 2 2 3 3
3 2 3 2 3
3 2 3 3 2
3 3 2 2 3
3 3 2 3 2
3 3 3 2 2

Hence, the no. of required numbers = 6

53. Consider the following statements:

1. The sum of 5 consecutive integers can be 100.
2. The product of three consecutive natural numbers can be equal to their sum.

Which of the above statements is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans. (c) : For, the sum of 5 consecutive integers can be 100 :

$18 + 19 + 20 + 21 + 22$
$= 100$

For, the product of three consecutive natural numbers can be equal to their sum :

$1 \times 2 \times 3 = 1 + 2 + 3$
$6 = 6$

Hence, Both 1 and 2 are correct.

54. A cubical vessel of side 1 m is filled completely with water. How many milliliters of water is contained in it (neglect thickness of the vessel)?

(a) 1000 (b) 10000
(c) 100000 (d) 1000000

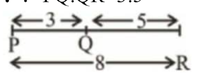
Ans. (d) : ∵ 1 m × 1 m × 1 m = 1000 liters

$1\ m^3 = 1000$ liters
$1\ m^3 = 1000 \times 1000$ milliliters
$1\ m^3 = 1000000$ milliliters

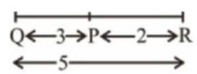
55. There are three points P, Q and R on a straight line such that PQ : QR = 3 : 5. If n is the number of possible values of PQ : PR, then what is n equal to ?

(a) 1 (b) 2
(c) 3 (d) 4

Ans. (b) : ∵ PQ:QR=3:5



PQ : PR = 3 : 8



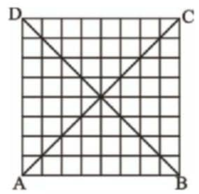
PQ : PR = 3 : 2

The number of possible values n =2

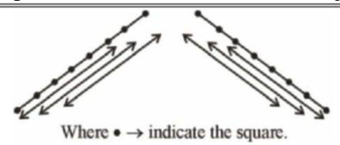
56. On a chess board, in how many different ways can 6 consecutive squares be chosen on the diagonals along a straight path?

(a) 4 (b) 6
(c) 8 (d) 12

Ans. (b) :



∵ any diagonal of chess board has 8 consecutive squares.



Where • → indicate the square.

thus, each diagonal less 3 ways.
Hence, the number of total ways is 6.

57. In an objective type test of 90 questions, 5 marks are allotted for every correct answer and 2 marks are deducted for every wrong answer. After attempting all the 90 questions, a student got a total of 387 marks. What is the number of incorrect responses?

(a) 9 (b) 13
(c) 27 (d) 43

Ans. (a) : Let the number of incorrect responses = x
∴ the number of correct answer = (90 – x)
According to question,

$(90 - x)(+5) + (x)(-2) = 387$
$450 - 5x - 2x = 387$
$450 - 7x = 387$
$7x = 450 - 387$
$7x = 63$
$x = 9$

Hence, the number of incorrect responses = 9

58. Consider the following addition problem :
3P + 4P + PP + PP = RQ2; where P, Q and R are different digits.
What is the arithmetic mean of all such possible sums?

(a) 102 (b) 120
(c) 202 (d) 220

Ans. (c) : 3P + 4P + PP + PP = RQ2
For finding the unit digit 2 in given sum
When put P = 3
33 + 43 + 33 + 33 = 142
When put P = 8
38 + 48 + 88 + 88 = 262
The arithmetic mean of all such possible sums

$= \frac{142 + 262}{2}$
$= \frac{404}{2}$
$= 202$

59. Consider the following multiplication problem: (PQ) × 3 = RQQ, where P, Q and R are different digits and R ≠ 0.
What is the value of (P + R) ÷ Q ?

(a) 1
(b) 2
(c) 5
(d) Cannot be determined due to insufficient data

Ans. (b) : It is possible only when

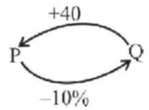
$(PQ) \times 3 = R\ Q\ Q$
$(85) \times 3 = 2\ 5\ 5$

P, Q, R are different digits R ≠ 0
∴ P = 8, Q = 5 R = 2
then (P + R) ÷ Q
$(8 + 2) \div 5 \Rightarrow 10 \div 5 = 2$

60. P scored 40 marks more than Q in an examination. If Q scored 10% less marks than P, then how much did Q score?

(a) 360 (b) 380
(c) 400 (d) 420

Ans. (a) :



Let scored marks by Q = x
then, scored marks by P = (x + 40)
∵ Q scored 10% less marks than P.

$\text{Required \%} = \frac{40}{(x+40)} \times 100\%$

$10\% = \frac{40}{(x+40)} \times 100\%$

$x + 40 = 400$
$x = 360$

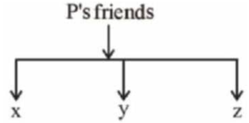
Hence,

Q scored 360 marks.

61. A person P asks one of his three friends X an to how much money he had. X replied, "If Y gives me Rs. 40, then Y will have half of as much as Z, but if Z gives me Rs. 40, then three of us will have equal amount." What is the total amount of money that X, Y and Z have?

(a) Rs. 420 (b) Rs. 360
(c) Rs. 300 (d) Rs. 270

Ans. (b) :



According to question,

$(x + 40) \qquad (y - 40) = \frac{z}{2}$

$Z = 2y - 80$ ..... (i)

again

$(x + 40) = y = (Z - 40)$
$\therefore x + 40 = y$ ……. (i)
$z - 40 = y$ ……. (iii)

From eq. (i) and (ii)

$z - 40 = y$
$2y - 80 - 40 = y \qquad \{\because z = 2y - 80\}$
$y = 120$

Put the value of y in the eq. (iii)

$z - 40 = y$
$z - 40 = 120$
$z = 160$

Put the value of y in eq. (ii)

$x + 40 = y$
$x + 40 = 120$
$x = 80$

Hence, $x + y + z \Rightarrow 80 + 120 + 160$
= ₹ 360

62. What is the value of 'X' in the sequence 2, 7, 22, 67, 202, X, 1822 ?

(a) 603 (b) 605
(c) 607 (d) 608

Ans. (c) :

2 , 7 , 22, 67, 202, X, 1822
2×3+1 7×3+1 22×3+1 67×3+1 202×3+1 X×3+1

∴ X = 202 × 3 + 1 and 607 × 3 + 1
X = 606 + 1 1821+ 1
X = 607 ⇒ 1822

63. Half of the villagers of a certain village have their own houses. One fifth of the villagers cultivate paddy. One-third of the villagers are literate. Four-fifth of the villagers are under 25 years of age. Which one of the following statements is certainly correct?

(a) All the villagers who have their own houses are literate
(b) Some villagers under 25 years of age are literate
(c) Only half of the villagers who cultivate paddy are literate
(d) No villager under 25 years of age has his own house

Ans. (b) : ∵ Fractions $\Rightarrow \frac{1}{2}, \frac{1}{5}, \frac{1}{3}, \frac{4}{5}$

and denominators of factions = 2, 5, 3, 5
∴ LCM of (2, 5, 3, 5)
LCM = 30
∴ Let total villagers = 30